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Micro and Macro Databases

- Dr. Rahul S.

The Digital Skills Paradox: Access, Capability, and Utilization Gaps in India's Internet Economy

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Abstract

Investigating the "digital skills paradox" in India, this research reveals that there is a concurrence of extensive internet accessibility and excellent digital manpower, but the skills remain largely untapped. The study based on individual level data extracted from the NSSO 80th Round Modular Telecom Survey (2025) investigates the difference between the digital capability and the actual use of Indian internet users. The authors through descriptive and multivariate regression techniques, identify five interconnected paradoxes: urban, access-quality, usage-frequency, device, and skill-type, that confront linear models of digital inclusion. The results indicate that 8.26% of digitally skilled people have reported unutilized skills. Urban and daily internet users are the ones who face the greatest capability-utilisation gaps. The paper asserts that both structural and contextual factors instead of solely social identity limit meaningful digital participation. The policy implications propose that a transition should be made from access- and training-focused initiatives to the development of digital application ecosystems that would bind the skills with the real-world opportunities.

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Introduction

India's digital transformation is often regarded as a huge success, and it is also the case that the country's rapid mobile internet expansion, lower data prices, and increased state-sponsored digital infrastructure were the main contributors to the change. Today, digital technologies permeate almost every aspect of life - governance, welfare, finance, education, and economic activity. Thus, the picture of almost complete digital inclusion is painted. Nonetheless, this story conceals the appearance of new forms of digital inequality.

On the whole, digital access is extensive but still, many people do not take advantage of the connectivity and therefore do not use the internet to its full potential in a meaningful way. Digital poverty is now indicated by the factors of persistent rural–urban gaps in the utilization of services, uneven usage patterns, and constant reliance on poor-quality infrastructure. As policy attention has gradually shifted toward digital skills, a paradox has arisen: having digital skills does not guarantee their frequent or effective use. A lot of people claim they are capable of using digital tools, but they do not make use of that capability, indicating that the traditional linear digital inclusion models are limited in their assumptions, which is that access leads seamlessly to capability and use.

The Digital Skills Paradox: Capability Without Practice

One of the central paradoxes of the Indian internet economy is that the digital skills have been turned into dormant potential rather than being put into practice. The capabilities of the individual user are often growing faster than the conditions that provide the possibility of regular and meaningful use. In addition, the engagement of users with the internet is limited to a very small part of the internet as factors such as poor quality of access, shared or mobile-only devices, cost concerns, fear of fraud, and limited control over digital spaces come into play. Thus, the connected users are left with skills that are underutilised. These patterns, on the one hand, show that digital agency is not only an individual capability but also a combined effect of the technological and the social environments in which digital interaction takes place.

Research Focus and Objectives

The current research unfolds against the backdrop of what can be termed the *digital skills paradox in India: the coexistence of great digital access and proclaimed digital competency along with the constant technical skill non-application in daily life*. Instead of considering digital inclusion purely as a matter of being connected or having the required skills, the study stresses the conditions under which digital capabilities are transformed or not transformed into practical use.

The main question that directs this investigation is:

In what way do the factors of household access, digital skill and usage context vary and contribute to the continuity of the capability-utilisation gap in India, even among active internet users?

The research has three main goals, which are very closely interrelated. To begin with, it investigates the reasons behind the digital skills of people who do not use them at all or only to a very limited extent, and it distinguishes between basic and advanced digital use in this respect. Furthermore, it looks into the issue of gender, caste, age, and geographical location as factors that cause the gap between technology access and usage to continue prevailing. Finally, it questions if the quality of access improvement leads to the same proportion of digital use or if, on the contrary, a paradox is set up where the potential is expanded faster than the actual application. In this respect, the study reflects on digital engagement as the capacity to turn digital potential into significant participation and developmental outcomes rather than being merely an issue of connectivity and ability.

Literature Review

The digital divide has transformed from an issue of access that was binary to a multi-dimensional phenomenon that included not only skills but also usage and outcomes. The initial works that has the "first-level digital divide" as a reference point were mainly concerned with the physical access to technology and the state of connectivity[1]. Global access helped to stimulate research that moved towards identifying the "second-level divide," which drew attention to the differences in digital skills and literacy as the main problems[2]. In the Indian situation, there have been considerable findings through research revealing that there are still major gaps existing in access to the internet as well as in the basic digital skills of people from different rural and urban areas, of different genders and castes (Chakravarty & Mandal, 2020).

The recent critiques of the linear model of "access → skills → use" indicate that the mere mastery of the skills does not reach to their regular or meaningful application. The capability approach of Sen is more frequently used in the context of digital inclusion, and this approach views it as not having resources but rather as the power to realize one's capabilities that are valued. This view points out the "conversion factors", namely, the social, institutional, and personal conditions that make it possible to turn access and capability into real use (Kleine, 2013). In India, research findings show that mobile-only access, device-sharing, and gendered limitations often prevent the digital literacy from being turned into a consistent practice (NSSO, various rounds).

An increasing number of researchers point to the existence of a "usage gap" or "third-level divide" where people have access to the internet and possess the skills but still do not participate in online activities that are beneficial or empowering (Wei et al., 2011). This issue is mainly observed in countries like India, where the spread of mobile internet is very fast, but the use contexts and application ecosystems are not developed accordingly. Researchers say that policy

[1] Dan Deursen, A. J., & van Dijk, J. A. (2014). The digital divide shifts to differences in usage. *New Media & Society, 16*(3), 507–526

[2] Hargittai, E. (2002). Second-level digital divide: Differences in people's online skills. *First Monday, 7*(4)

measures like Digital India often give more importance to infrastructure metrics than to people's actual engagement, thus there is a risk of creating a population that is "digitally skilled but inactive" (World Bank, 2021).

This literature review places the present research at this decisive moment, not merely mapping access and capability gaps but also probing the paradox of digital skills being little used by the connected Indian population. It is founded on the capability approach to analyze the systemic and contextual constraints that inhibit the transfer of gained skills to ordinary practice, thereby tackling a major gap in comprehension of India's digital transformation.

Data and Methodology

Data Source

This research paper utilizes the unit-level data of **NSSO 80th Round Modular Telecom Survey**, which was done in year 2025. The survey is among the foremost and very large nation-wide representative sets of data regarding digital access, internet usage, and digital skills in India. It makes use of a stratified multi-stage sampling design that comprises all states and union territories and includes both rural and urban areas.

The survey consists of two interconnected datasets: **a household-level schedule (CMST80HH)** and a **person-level schedule (CMST80PER).** The household dataset encapsulates data on internet accessibility, type of connection, and socio-economic characteristics, whereas, the person-level dataset logs individual internet-use, digital skills, frequency of use and device type. The presence of shared identifiers enables the joining of household and individual records; thus, the study can investigate how household access conditions affect an individual's digital outcomes.

Strengths and Limitations

The NSSO 80th Round's major strength is that it not only indicates the internet access in binary terms but also shows different skills of the users and their recent use in detail. Different groups like gender, caste, income, and rural versus urban location can be analyzed in detail thanks to the large sample and national coverage. The survey's distinguishing between the ability to perform digital tasks and the actual performance is very important for the examination of gaps in capability–utilisation.

On the other hand, the dataset has some disadvantages. Digital skills and usage are reported by the respondents themselves and hence may be affected by recall or reporting bias issues. The survey period indicates whether or not skills were used, but it does not provide information

about the intensity or the quality of usage. Furthermore, the cross-sectional nature of the data limits the causal interpretations that can be made. These limitations have been accepted while drawing the conclusions.

Variable Selection and Data Construction

Household-Level Variables (CMST80HH)

The household dataset was used to choose variables that represent the socio-economic status and the access quality of the households:

- The **Monthly Per Capita Consumption Expenditure (MPCE)** was divided into quintiles showing the economic position of the household.
- The **internet service at home** was marked as a binary variable.
- The **fixed-line internet connection** (like broadband or wired access) was marked separately to categorize the access quality to be higher than the mobile-only connectivity.
- The **religion** and the **social group (caste)** were kept for the social stratification analysis.
- The **sector** (rural-urban) was used to identify differences in location.

These variables contributed to the formulation of a **Household Access Quality Index (HAQI)** which is explained below.

Individual-Level Variables (CMST80PER)

Digital capability, usage and demographic characteristics were specified through selection of variables from the person-level dataset:

- **Internet usage ability** and **Internet usage.**
- Digital skills comprising **email, online banking, copy-paste operations, document handling and presentation skills.**
- The **recent performance of selected skills** (copy-paste, email/messaging, presentations), which makes it possible to determine the skills that have not been used.
- **Gender, age, device type**, and **frequency of internet use**.

The individual dataset was combined with the household dataset by means of common identifiers, which made it possible to examine the effect of household access conditions on individual digital behaviour.

Construction of Digital Indices

Individual Digital Capability Score (IDCS)

The Individual Digital Capability Score (IDCS) was derived as an additive index representing the wide range of digital capabilities. The index adds up binary indicators for digital

engagement, internet usage, and five main digital skills. The higher the values the more digital capability exists. The index shows digital capacity that is potential rather than actual utilization.

Household Access Quality Index (HAQI)

The Household Access Quality Index (HAQI) reflects the qualitative differences in access and not just the availability. This index consists of three elements: internet availability in the home, the presence of a fixed connection, and computer usage. The index classifies access quality from low to high, thereby marking the boundaries between constrained, mobile-dependent households and those enjoying more liberating digital environments.

Capability-Utilisation Gap (CUG)

A Capability-Utilisation Gap (CUG) was established as the central outcome of interest measurement. For the chosen skills, persons who acknowledged having a skill but had not exercised it lately were marked as having a capability that was not in use. Such gaps were added up to create an individual-level CUG score denoting the digital skills that were known but not used. The underutilisation of digital capability is reflected by higher scores.

Analytical Strategy and Models

The analysis is carried out in three steps.

First of all, **descriptive analysis** was applied to understand differences in digital capability (IDCS) and capability-utilisation gaps (CUG) by caste, gender, income brackets, and rural–urban divide.

Next, **logistic regression models** were applied to binary outcomes, like whether a person is an internet user or whether a certain digital skill is still laying idle. In such models, the demographic characteristics, socio-economic status, and access quality are taken into account simultaneously to predict the likelihood of using the digital service while keeping other factors constant. Odds ratios are provided to explain the strength and direction of associations.

Finally, **linear regression models** were the tools of choice for analyzing continuous outcomes like IDCS and CUG. The focus was on how digital potential and non-utilized competencies differ regarding gender, age, nationality, economy, sector, and quality of household access. The sequential models incorporate access quality, kind of device, and frequency of usage to investigate the extent to which social disparities remain after controlling for usage conditions.

Further requirements comprise:

1. Binary logistic models that spot persons having large capability-utilization gaps.
2. Multinomial models that tell apart various levels of CUG severity.
3. Interaction models that look into the joint contributions of caste, gender, device type, and location in determining utilization gaps.

4. Robustness checks that narrow down the sample to active internet users in order to see if utilization gaps still exist even among connected people.

These models, combined together, enable the research to investigate not only the access and skills but also the structural and contextual reasons that keep digital capabilities undervalued in India.

Results

The Three-Tiered Digital Divide: Access, Capability, and Utilization

Digital Capability Divide Across Social Groups and Sector

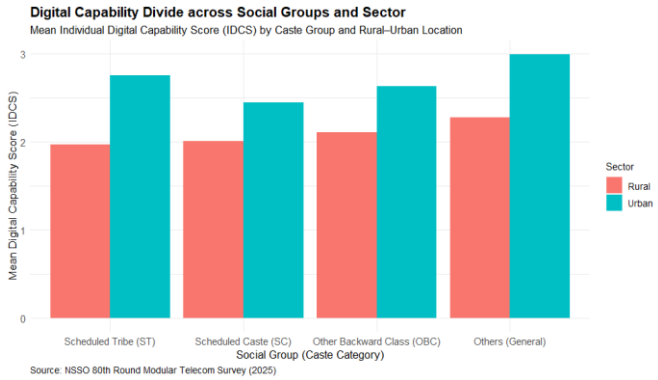

Figure 1

The analysis has made it evident that there is a significant digital capability hierarchy among different social groups and sectors. The individuals living in urban areas have always had a much better Individual Digital Capability Scores (IDCS) than those in rural areas, which is true for all castes, thus confirming the ongoing urban skill acquisition advantage. A social gradient has been established in which Scheduled Tribes are at the bottom of the capability levels (Rural: 1.97; Urban: 2.76) and Others/General category is at the top (Rural: 2.28; Urban: 3.00).

The urban-rural divide is clearest in the case of the Scheduled Tribes, suggesting that the tier of location is a significant factor for the most disadvantaged groups. Moreover, the increase in digital capability does not lead to a corresponding increase in usage across all groups, as later results indicate that groups with the highest capabilities also have the largest gaps between the skills they possess and the skills they actually use, this is the new digital skills paradox which has been highlighted.

Social and Spatial Patterns in the Capability–Utilisation Gap

It was not expected that the capability–utilisation gap would show such a slight difference across the wealth of social groupings. Scheduled Tribes and Scheduled Castes have about 0.06 unused skills as an average, while OBC and Others/General groups have only a little higher gaps of about 0.07. The distribution of the cumulative gap among groups is relatively flat, which is in contrast to the steep social hierarchy recognized in the digital capability itself, and it implies that the gap in usage is not strongly influenced by the caste system even though the digital skills are acquired unevenly.

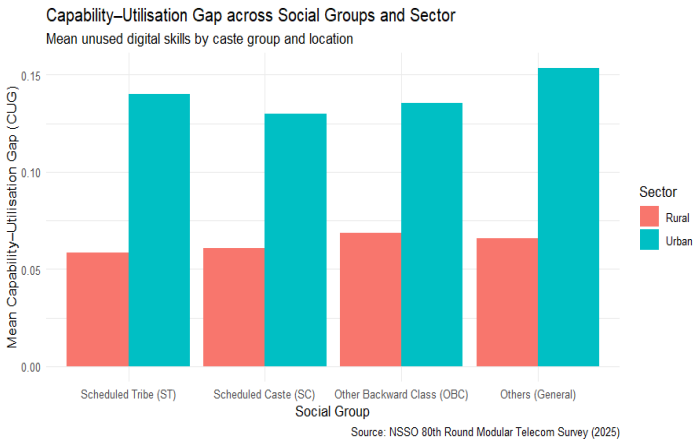

Figure 2

This situation creates a clearer paradox when the differences between sectors are considered. Although urban residents have better access and have developed higher digital capability, their average capability–utilisation gap is about three times larger (mean CUG = 0.141) than that of rural residents (mean CUG = 0.0645). This means that the most privileged members of society, both in terms of space and social factors, have more digital skills but do not necessarily use them more often. This finding points to the fact that enhanced infrastructure and skill training by themselves are not enough to promote digital engagement; rather, they bring to the fore the very factors of context and structure that shape digital practice as barriers.

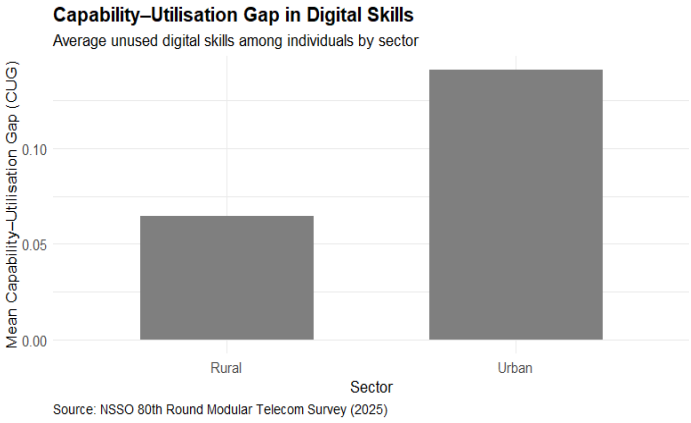

Figure 3

Distribution of the Capability-Utilisation Gap (CUG)

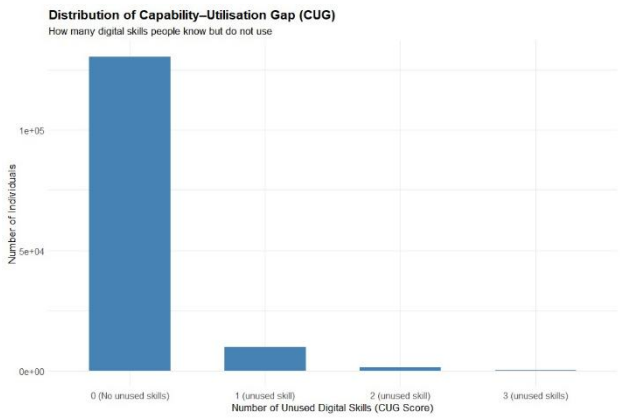

Figure 4

India's digitally skilled population is mostly engaged in using their skills but a small group reflects a discrepancy between what they can do and what they actually do. Around 8.26% of the skilled population reported that at least one of their digital skills is not being used. Among them, 5.1% have one skill that is not used, 2.4% have two skills that are not used, and 0.8% have all three skills but none of them are used. Despite the fact that a large number (91.74%) of the skilled workers make full use of their digital capabilities, the small group with constantly unused skills is nevertheless of great significance, as it points at the existence of structural and contextual barriers which inhibit some people from putting their digital knowledge into daily use.

Regression Analysis: Explaining the Paradox

Determinants of Digital Skill Underutilization (Linear Regression - Model A2)

Factor	What is observed	What it means in simple terms
Household Access Quality (HAQI)	Higher access quality is linked to more unused skills	Better internet and devices help people *learn* skills faster than they can *use* them
Urban residence	Urban users have more unused skills than rural users	Living in cities increases exposure to skills, but not always opportunities to apply them
Age	Older individuals show slightly higher gaps	Skills accumulate over time, but usage does not keep pace
Daily internet use	Daily users show the largest utilisation gaps	Heavy exposure creates skills, but many remain unused
Computer use (vs mobile-only)	Computer users have fewer unused skills	Computers make it easier to actually apply digital skills
Gender	No meaningful difference	Men and women show similar patterns of skill underutilisation

[3]

The table below illustrates the principal digital skills paradox recognized in this research. When it comes to technology usage, the opposite happens, as more use and better internet infrastructure together mean more unused digital skills rather than less. The people living in

[3] This table explains Why Digital Skills Are Not Used?
Sample size: 142,065 individuals
Overall fit: The model explains meaningful but not complete variation, indicating structural factors beyond individual traits.

the city and using the internet every day, who are usually viewed as having digital advantages, have the most significant difference between the skills they have and the skills they use actively.

However, at the same time, with computer access, the gap in the skills is reduced; this indicates that the nature of the digital environment plays a more significant role than access. Gender differences are reported to be small, suggesting that underutilised skills are not so much a result of individual identity as they are of structural and contextual barriers. Digital inclusion, as seen from the results, might only be a partial solution if it is limited to providing access and connectivity merely.

Predicting Extreme Underutilization (Logistic Regression - Model B)

This model analyses the elements linked to the high-level under-utilisation of digital skills, which is defined as the situation where people have digital skills but do not use any of them at all.

One of the most interesting results is that the daily internet users are nearly twenty times more likely to fully underutilise their skills, which is a clear indication of the digital skills paradox as those who are regularly exposed do not use the skills meaningfully. The urban dwellers also have a significantly higher chance of not using the skills completely which could be interpreted as the cities' better infrastructures and connectivity accelerating the skill acquisition faster than the opportunities for application.

Digital practice limitations reflected through women's experience of full underutilisation which is twice more than for men indicates that merely providing equal access does not solve the problem of gendered constraints.

Scheduled Caste individuals are nearly twice as likely as Scheduled Tribes to exhibit extreme gaps, a fact that points to the continued role of structural disadvantage in the digital divide issue, even when basic access is provided. While better household access quality boosts the chances of non-use, higher income comes with only limited protection. All in all, these results convey that those who are most connected to digital systems tend to be the least capable of transforming their digital skills into actual use, which in turn reinforces the main argument that there is capability without practice.[4]

The discrepancy in the use of digital skills is very pronounced when it comes to skill types. Basic skills like email and copy-paste show almost no non-use (roughly 5.5%) while almost 23% of the people who have the advanced skill of presentation software do not use it at all. This stark difference points towards a learning-application mismatch where advanced digital skills are learned but there are not enough everyday contexts to use them. The trend supports the research's main point that increasing digital capability does not necessarily lead to significant or continuous digital practice.

[4] The Extreme Paradox: Those most connected (daily users) are 19.9 times more likely to possess skills they never use. This suggests digital saturation without application.

The Device × Location Interaction (Model E)

This model investigates the effect of device type on digital skill underutilisation and the rural-urban divide simultaneously. It unveils the fact that the urban gap in usage is primarily due to mobile-only access, whereas the access to a computer allows for the meaningful application of skills in the urban settings.

Factor	Effect on CUG	What it means
Computer × Urban	−0.156 ***	Urban computer users have much lower unused skills
Mobile × Urban	+0.029 **	Urban mobile-only users have slightly higher unused skills
Computer user	+0.186 ***	Computer users have higher overall digital capability
Urban (overall effect)	+0.045 ***[5]	Urban areas show higher underutilisation on average

The urban penalty in digital skill utilisation appears to differ depending on the device used. It seems that urban living typically leads to the situation where, the digital skills are not used to the full extent, but this situation does not apply to computer users, who are more prone to making use of their skills, the opposite is the case. Urban mobile-only users, on the other hand, show that the digital skill gaps are even wider, which points to the fact that it is the device environment and not the access alone that determines whether digital proficiency is put into practice or not.

Multinomial Regression - Different Gaps, Different Determinants (Model C)

This multinomial regression analysis puts individuals with different levels of digital skills underutilisation (i.e., low, medium, and high gap) in contrast to those with no gap. The study indicates that various factors are of different significance at different levels of underutilisation, rather than one single cause being responsible for all gaps.

What Increases the Risk of Skill Underutilisation

Predictor	Low Gap (1 skill unused)	Medium Gap (2 skills unused)	High Gap (3 skills unused)
Urban residence	1.85× higher risk	**2.46× higher risk**	1.97× higher risk
Better access (HAQI)	**1.96× higher risk**	1.13× higher risk	1.78× higher risk
Female	Slightly lower risk	Slightly higher risk	Higher risk
Richest income group	No clear effect	**Lower risk**	Lower risk

The study indicates that being in an urban area has a substantial impact on the risk of medium-level underutilisation, i.e. non-usage of skills is mostly relegated to cities. On the other hand, access quality (HAQI) has its major impact in the case of small gaps, where very well-connected people are more likely to leave at least one skill unused. All three skills being unused (severe underutilisation) is a scenario that occurs both in urban areas and for people with access

[5] ***$p < 0.001$, *$p < 0.01$

to higher quality resources, thus strengthening the paradox that better resources do not necessarily equal usage. Altogether, the results indicate that various degrees of underutilisation are caused by different factors, not just one digital divide.[6]

Adjusted Predictions

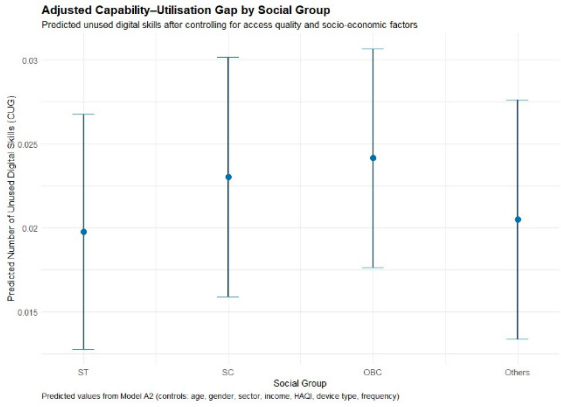

Figure 5

[7]Figure 5 illustrates the predicted capability–utilisation gap (CUG) across social groups after differences in age, gender, rural–urban location, income, household access quality (HAQI), device type, and internet usage frequency were considered. When these variables are controlled for, the average number of redundant digital skills is quite similar, not varying much at all, being between 0.02 and 0.03 only. There is a great overlap in the confidence intervals which implies that no statistically significant differences in the use of digital skills remain among the caste groups.

This result leads to a conclusion that the disparities in digital skill use among social groups observed are not attributed to caste identity per se, but rather to unequal access, different device usages, and different levels of digital engagement. To put it differently, the more social groups are digitally contextually comparable the more the chances of them leaving skills unused become. This emphasizes the main point of the research: the Indian digital skill paradox is less attributed to social identity and more to the existing structural and technological conditions that influence how digital competencies are actually used in daily life.

The Access-Utilisation Paradox

The paper reveals a fascinating and somewhat opposite connection between the quality of household access and the application of digital skills. Individuals from households with the best digital access, after considering factors like age, gender, caste, income, sector, type of device, and internet usage, show more significant capability–utilisation gaps. When the Household Access Quality Index (HAQI) moves from 0 (poor access) to 3 (high access), the number of predicted unused digital skills increases from about 0 to almost 0.06.

[6] Reported values are relative risk ratios from a multinomial logistic regression, comparing each gap category to individuals with no capability-utilisation gap.
[7] Adjusted predicted values from Model A2 with 95% confidence intervals.

[8]The positive trend suggests that one more unit rise in HAQI is linked to an increase of about 0.03 in the number of unused skills. This indicates that the households having the best connectivity and devices are more likely to acquire digital skills that are not used on a continuous basis. Instead of lessening under-utilization, it seems that upgrading access has a part in skill acquisition without having the meaningful use of the skill, thus increasing the paradox of the study: access and capability emergence alone do not assure digital engagement.

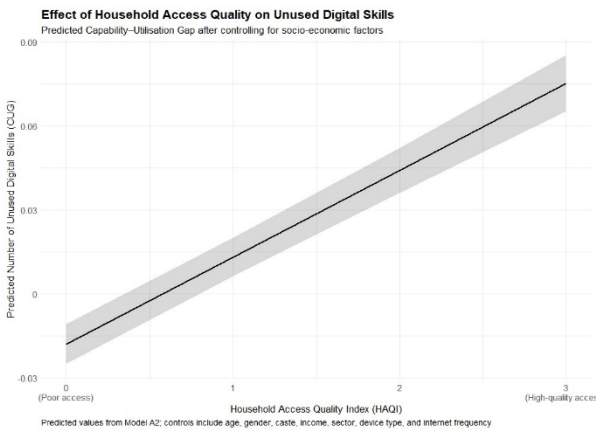

Figure 6

The entire analysis reveals various interrelated paradoxes that question the traditional ideas of digital inclusion.

The first one is the urban paradox which states that even if the residents of the urban areas have much better digital capabilities, they have also, at the same time, very large gaps between the skills they have and the skills they use.

The second paradox is access, which reveals that the better the quality of household access, the more underutilization of skills rather than improvement in digital practice.

The third paradigm reflects the usage frequency paradox and daily internet users stand as the ones most probably to completely underutilize their skills, hinting that continuous connectivity may not be equal to productive or functional use.

Fourth comes the device paradox which says that urban disadvantage in consumption disappears for people using computers but is still there for mobile-only users, thus proving how important the environment of usage is.

Lastly, the skill-type paradox states that advanced digital skills are much more likely to be left unused than basic ones, thus signaling the mismatch between skill acquisition and chances for application. These contradictions expose the **main conclusion of the research: the digital divide in India is not just about access or support anymore, but rather about the actual practicing conditions of the digital skills.**

[8] The solid line represents predicted values from the multivariate regression model, while the shaded band indicates the 95% confidence interval around these predictions. All estimates are adjusted for age, gender, caste, income, sector, device type, and internet usage frequency.

Discussion

This research advances the traditional digital divide analysis by revealing that having access to and being trained in the use of technology do not necessarily result in the use of technology that is meaningful. The outcomes uncover consistent discrepancies between digital power and its usage, which underline a group of contradictions with significant consequences for both theory and policy.

Interpreting the Paradoxes

The **urban paradox** illustrates that urban people, even though they have higher digital skills, still have a much larger gap in utilization. This indicates that higher exposure and availability do not ensure productive use, probably because of the competing activities that take up time and the predominance of non-productive digital activities in city areas.

The **access-quality paradox** is a further confirmation of the difference between resources and outcomes: families with better internet access also show higher levels of unused skills. This is in line with Sen's notion that resources can create possibilities but do not necessarily convert them into functionings unless there are favorable social and institutional conditions.

The **usage-frequency paradox** is yet another challenge to the often held beliefs. Internet users who are online daily are very likely to completely underutilise their skills, which points out to digital saturation without meaningful engagement.

The **device paradox** demonstrates that the penalty experienced by urban dwellers vanishes for those who use computers but continues for those who rely solely on mobile phones, illustrating that the type of device used determines whether skills are put to use or not. Last but not least, the **skill-type paradox** indicates that highly trained skills are much more likely to go unused than basic ones, which is due to the mismatch between the training provided and the opportunities for application in the real world.

The results of these studies, taken together, throw doubt on the linear models that predict a one-way process of:

Access → Skills → Use

Digital inclusion was rather to be regarded as dependent on the application ecosystems comprising workplaces, institutions, and social contexts that offer and require digital practice.

The research, besides adding to the digital divide theory, also empirically showed the existence of a third layer of inequality, application besides access and capability. Moreover, it gives a prominent place to the conversion factors, which include device type, usage context, and opportunity structures, in the process of translating skills into economic and social outcomes.

Micro-Macro Connections and Policy Context Analysis

The individual level results of the study on digital skill underutilization aggregate to considerable macro-level economic and social impacts. For example, 8.26% of the digitally skilled Indians represent the "unrealized functionings," meaning they have capabilities but do not apply them, which makes them resources that do not create worth[9]. Scaling this up to the whole country means that there are millions of digital workers who are not using their skills and thus, have a direct negative effect on the productivity growth in India. The World Bank (2021) says that if digital skills were used effectively, it could result in 1-1.5% increase in India's annual GDP; our study indicates that this potential is only partially realized because of application gaps and not because of lack of skills.

The household access paradox is telling us that the digital investments that are made at the national level are not very efficient. India's digital investments under the Digital India program since 2015 have reached around ₹1.5 trillion[10], but still households with the highest access quality (HAQI=3) have 0.06 more unused skills than those with the lowest access. This is the case being made for the suboptimal return on infrastructure investments - the money spent on connectivity is not producing the same amount of productive digital engagement as before. The positive correlation between access quality and skill underutilization ($\beta=0.031$, $p<0.001$) indicates that the existing policies might be creating digital capability quicker than the application ecosystems can take in.

The use of social patterns in skills among groups is one of the factors that bring about the most significant changes in terms of equity in the digital economy of India. The application of traditional caste hierarchies in the acquisition of digital capabilities is shown in the first figure (1), while our adjusted predictions (Figure 5) reveal that when access and context are controlled, the application gaps become equal. Thus, it is found that it is structural barriers that limit digital participation rather than social identity. In India, digital inequality is more a reflection of the existing socio-economic disparities than a factor creating new ones.[11] Moreover, our findings show that even in the case of skills being acquired, the same structural factors still constrain the application.

The interaction between device and location has ramifications on infrastructure policies that are direct as well as digital. Urban computer users have a comparably smaller extent of underutilization (-0.156, $p<0.001$), while urban mobile-only users have the largest differences. This indicates that the ecology of devices is more important than the connectivity by itself. The current policies that focus on mobile penetration (with countries like India having some of the lowest data prices in the world) might not be enough to foster digital engagement in a productive way. The National Digital Communications Policy 2018 predicted that 50% of households would have broadband access by 2022; our results indicate that such a policy

[9] Sen, A. (1999). *Development as freedom*. Oxford University Press
[10] Government of India. (2022). *Digital India: Annual Report 2021-22*. Ministry of Electronics and IT
[11] Chakravarty, S., & Mandal, K. (2020). Digital divide in India: An empirical study. *Economic and Political Weekly, 55*(45)

should be coupled with initiatives that enhance the ecosystem of productive devices, especially for the disadvantaged groups.

Policy Context: Measuring What Matters

The present Digital India monitoring system chiefly measures access through the metrics of connections, transactions, and platform registrations. While the Digital India Dashboard indicates more than 1.2 billion Aadhaar enrolments and 450 million UPI transactions per month, it does not provide any metrics for skill application quality or productive engagement. Likewise, Skill India also evaluates its success based on the number of trained individuals rather than looking at post-training skill utilization, with over 50 million certified since 2015.

This discrepancy in the measurement is a "second-level digital divide" in policy evaluation: One measures the first barrier (access/training) but not the second (meaningful application). [12]Our CUG metric presents a possible remedy - a commonly accepted measure of digital skill application that could be integrated into the national monitoring frameworks.

The one thing that is not there in India's digital policy framework is application opportunity creation. Although PMGSY (Pradhan Mantri Gramin Digital Saksharta Abhiyan) has made it possible for more than 60 million people to become digitally literate, no such or similar contexts in which these skills could be applied have been created yet. Successful international models like **Singapore's SkillsFuture** which combines training with application in projects at the workplace, imply possible solutions. What India requires is **'Digital Application Ecosystems'** as we put it, interconnected systems that link skill acquisition with opportunities for significant use in jobs, governance, and community development.[13]

Future Research Directions

1. **Longitudinal Panel Studies**: Monitoring the same people for 3-5 years to find out if and when digital skills that are never used become activated, lost, or remain forever dormant.
2. **Mixed-Methods Deep Dives:** Use a combination of skill surveys, in-depth interviews, and digital ethnography to get the "why" behind non-application—cognitive, motivational, social, and structural barriers will be explored.
3. **Policy Experiments (RCTs):** Create and evaluate the effect of real-world digital interventions like community workspaces, project-based learning modules, or tech mentorships to find out what is the best way to turn skills into practice.
4. **International Comparative Analysis:** Compare India's capability utilization gaps with those of other major emerging economies (Brazil, Indonesia, South Africa) in order to extract policy lessons and identify contextual factors.

[12] Hargittai, E. (2002). Second-level digital divide: Differences in people's online skills. *First Monday, 7*(4)
[13] World Bank. (2021). *Digital India: Technology to transform a connected nation*. World Bank Group

5. **Skill Application in Specific Domains:** Start with a research study to find out how digital skills are used (or not) in education, healthcare, agriculture, and gig work to plan application strategies targeted to specific sectors.

CONCLUSION & POLICY IMPLICATIONS

This research uncovers one of the most important and yet least studied factors of India's digital transformation: the huge underutilization of the digital skills that have been acquired. Quantitatively, 8.26% of people with digital skills report having skills that they do not use, and this is especially true for city-dwellers who have a gap in the use of their capabilities that is 118% higher than that of people living in the countryside. The most amazing fact is that regular internet users are 19.9 times more likely than non-users to completely underutilize their skills, while advanced skills such as using presentation software have a 4 times higher non-application rate than basic skills.

These results can be grouped into five closely linked paradoxes that question the concept of digital inclusion applied in a linear way: the Urban Paradox, the Access Paradox, the Usage Frequency Paradox, the Device Paradox, and the Skill-Type Paradox. They all together emphasize that the journey from digital access to digital capability and then to active digital practice is not a matter of course nor a matter of luck.

Policy Recommendations: A Three-Tiered Roadmap

In order to move inactive digital human capital into the area of productive engagement, it is necessary to have a staged, evidence-based policy approach.

- **Immediate Actions (1-2 years):** First, the national monitoring frameworks should be revised. The Digital India Dashboard must go beyond monitoring access and transactions and include metrics of skill application quality, such as the Capability-Utilisation Gap (CUG) developed here.[14] The second point is to start implementing device-specific interventions. The analysis we carried out has revealed that mobile-only access causes contextual disadvantage; so, subsidies or public provisioning of computers/tablets, especially for rural students and women, can help to mitigate this disadvantage.
- **Medium-term Strategies (3-5 years):** To transform digital literacy programs from being certificate-focused to project-based application, initiatives like PMGDISHA should integrate modules where learners directly apply their skills to the community needs, like making local business inventories or digital archives[15]. At the same time, develop "Digital Opportunity Portals" that link professional workers with micro-tasks,

[14] Government of India. (2022). *Digital India: Annual Report 2021-22*. Ministry of Electronics and IT
[15] World Bank. (2021). *Digital India: Technology to Transform a Connected Nation*. World Bank Group

civic tech initiatives, or gig jobs, thereby creating a connection between the learning and the doing.

- **Long-term Vision (5+ years):** Support integrated Digital Application Ecosystems. This means that there should be policy harmony among skilling, industry, and innovation ministries so that the skills training will be in tune with the changing market demands and will create real use contexts[16]. A national longitudinal survey should be set up to monitor over time the application of digital skills and at the same time a solid evidence base for future policy should be created. Last but not least, international learning frameworks should be institutionalized in order to adapt the successful models, like the Singapore's SkillsFuture ecosystem, to the Indian context.

Transitioning from a digital have mentality to that of a digital do is fundamental for India to achieve the full economic and social benefits of its digital investments. The present research forms the empirical basis for that significant change.

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